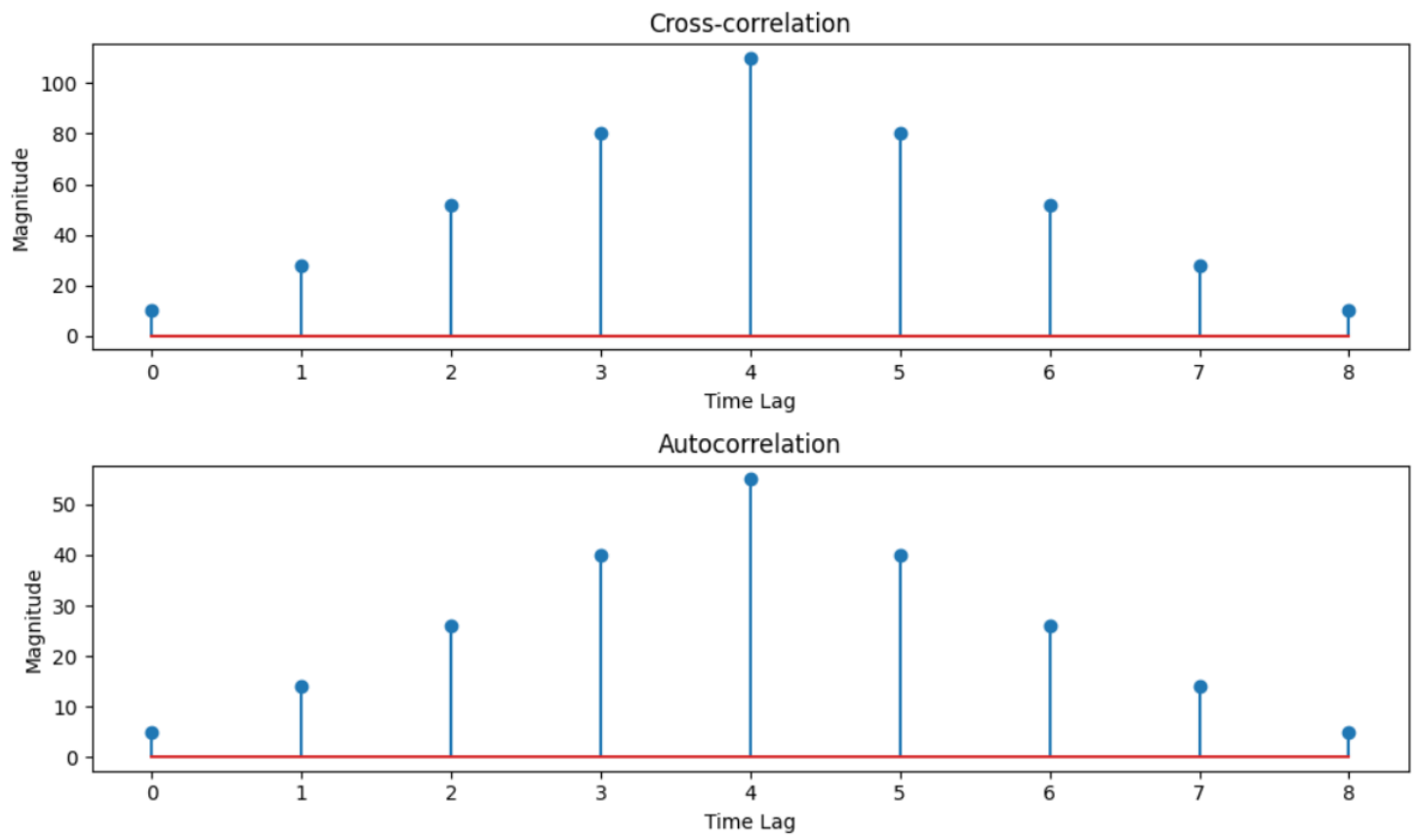


 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim: Simulate discrete time sequences.	
Experiment No: 2	Date:	Enrollment No: 92510133028

AIM: Simulate linear convolution and circular convolution on discrete time.

```
import numpy as np
import matplotlib.pyplot as plt
def cross_correlation(signal1, signal2):
    # Compute the cross-correlation
    cross_corr = np.correlate(signal1, signal2, mode='full')
    return cross_corr
def autocorrelation(signal):
    # Compute the autocorrelation
    auto_corr = np.correlate(signal, signal, mode='full')
    return auto_corr
# Define the discrete-time signals
signal1 = np.array([1, 2, 3, 4, 5])
signal2 = np.array([2, 4, 6, 8, 10])
# Compute the cross-correlation
cross_corr = cross_correlation(signal1, signal2)
# Compute the autocorrelation
auto_corr = autocorrelation(signal1)
# Plot the cross-correlation and autocorrelation signals
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
plt.stem(cross_corr)
plt.title('Cross-correlation')
plt.xlabel('Time Lag')
plt.ylabel('Magnitude')
plt.subplot(2, 1, 2)
plt.stem(auto_corr)
plt.title('Autocorrelation')
plt.xlabel('Time Lag')
plt.ylabel('Magnitude')
plt.tight_layout()
plt.show()
# Save the cross-correlation or autocorrelation signals (optional)
# np.savetxt('cross_correlation.txt', cross_corr, delimiter=',')
# np.savetxt('autocorrelation.txt', auto_corr, delimiter=',')
```

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Conclusion:

By completing this experiment, I gained a clear understanding of linear and circular convolution and their differences. I also learned how to implement these operations using Python and visualize the output signals with the help of NumPy and Matplotlib.